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CSA-101

Examination –May- 2024

B.Tech. I/II Sem : Common to all Branches

Introduction to Computer Science & Engineering

Time : 2 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:

Part a & b – Objective type/Fill in the blanks/One sentence answer.

Part c – Max 100 words and Part d – Max 400 words.

Word limit NOT to be followed for diagram, numerical, derivation.

			Marks	BL	CO
Q.1	(a)	_____ supplementary memory system that temporarily stores frequently used instructions and data for quicker processing by the central processing unit (CPU) of a computer.	02	BL1	CO1
	(b)	If a CPU receives input that says to complete a calculation or logical operation, this is done with the _____ unit.	02	BL2	CO1
	(c)	Briefly explain computer generations.	03	BL3	CO1
		OR			
		Compare System software with application software.	03	BL4	CO1
	(d)	Draw CPU components diagram & briefly explain it.	05	BL3	CO1
		OR			
		Give a diagram to show bus architecture and also give its importance.	05	BL3	CO1
Q.2	(a)	A _____ is a name given to the variable whose values can't be altered or changed.	02	BL2	CO2
	(b)	In C using _____ with a data type we cannot store negative values in it.	02	BL2	CO2
	(c)	Differentiate between if and switch statements.	03	BL4	CO2
		OR			
		Define Data type and give the list of data types in C.	03	BL2	CO2
	(d)	Differentiate between logical & bitwise operators and also write a C program to demonstrate it.	05	BL4	CO2
		OR			
		Write a program in C to find the area, diameter and circumference of a circle.	05	BL5	CO2

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Q.3	(a)	_____ is the scope of the variables, which are declared outside of all functions.	02	BL2	CO3
	(b)	_____ specifies the region of the program from where variable can be accessed.	02	BL2	CO3
	(c)	Give the types of recursion with example.	03	BL2	CO3
		OR			
		Define static and extern storage classes in C.	03	BL2	CO3
	(d)	Take an array of 10 integers and accept values into it. Sort the array in descending order.	05	BL3	CO3
		OR			
		Write a program to print Multiplication Table from 1 to 5.	05	BL4	CO3
Q.4	(a)	_____ are those structures that have one or more pointers which point to the same type of structure, as their member.	02	BL2	CO4
	(b)	When names of data types become difficult to use in programs, _____ is used with user-defined data types, which behave similarly to defining an alias for commands.	02	BL1	CO4
	(c)	Give reasons to justify the importance of #define.	03	BL3	CO4
		OR			
		Show how Union is different from structure.	03	BL3	CO4
	(d)	Explain file handling. Show how to write "Hello World in C" in a file and then read the contents of the file.	05	BL3	CO4
		OR			
		Discuss the operators used in structure with examples.	05	BL4	CO4
Q.5	(a)	The computers within a network may use vastly different software and hardware; however, the use of _____ enables them to communicate with each other regardless.	02	BL2	CO5
	(b)	_____ based protocols can be automated and decentralised, thus enabling the creation of cryptoassets without the need for controlling, supervisory or centralised bodies.	02	BL1	CO5
	(c)	Give examples to show various types of security in computers.	03	BL3	CO5
		OR			
		Differentiate between Machine learning and Artificial intelligence.	03	BL4	CO5

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(d)	Explain the principles of operating system.	05	BL3	CO5

OR

	Define Data science and explain where we need it.	05	BL3	CO5

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating

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